

Module 2 Project

Objectives:

Show evidence that you have learned the following course competencies by designing, implementing, and querying the database as described in this document –

- Create a database ERD while adhering to normalization rules
- Design database tables that adhere to normalization rules
- Deploy database tables that adhere to normalization rules
- Setup a ERDs using Vertabelo database design tools
- Create an ERD using Crow's foot notation
- Setup a MySQL database from an ERD document
- Setup primary keys in MySQL
- Setup foreign keys in MySQL
- Setup and execute queries in MySQL
- Create single-table database queries
- Create multi-table database queries using JOINS
- Complete self-evaluation and short reflection for Project 2 requirements

Project Description:

For the Module 2 project, you will be setting up a database to keep track of recipes and their ingredients. Reminder that a recipe can be made up of multiple ingredients and that each ingredient could be utilized within multiple recipes. As you design your tables and relationships, please write down any assumptions you are making if they are not clearly stated here already (or you think they might deviate from what is stated here).

The table below shows the columns you will need to track the necessary recipe and ingredient information. In this sample data table, there are 3 recipes and their corresponding ingredients listed. Note that the table is almost in 1NF, but the missing, duplicate data is omitted for this initial starting point. For this project, you should be normalizing the data to 3NF using the following known functional dependencies:

- (RecipeID, IngredientID) -> RecipeName, Notes, AcquiredFrom, IngredientID, Ingredient, Quantity, Technique
- (RecipeID) -> RecipeName, Notes, AcquiredFrom
- (IngredientID) -> Ingredient

You should add two more recipes to the sample data table. The recipes you add should share at least one ingredient between them. You will be using this sample data provided and that you added throughout this project.

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Your goal is to use this project description and sample data to design a database that is normalized to third normal form (3NF). This means that each table should be in 3rd normal form – once that is the case your database is considered normalized.

Draw out your normalized database design in Vertabelo utilizing Crow's foot notation. Once your database is normalized and designed, you should implement your tables, primary keys, and foreign keys in your MySQL DBMS schema that matches your username (e.g. cstudent, tgsmith27, etc.). Note that not all of the columns require data and be sure to indicate that in your table design (those columns are considered nullable). You can optionally utilize AUTO_INCREMENT for your primary key values or you can set them up manually. Also, choose appropriate data type designations for each of the columns in your tables.

After your tables and foreign keys are implemented, you should pick one (1) of the provided recipes and related ingredients to insert into your tables. Then also add the 2 extra recipes that were added by you as specified earlier in this document.

Finally, you should setup some SQL queries and save them in a *.sql file for submission with your project. You should run two (2) single-table SQL SELECT queries. You should run four (4) multi-table SQL SELECT queries. Multi-table means JOINing at minimum 2 tables together.

At some point in your six SQL SELECT queries you should utilize the following keywords at least one time:

WHERE

ORDER BY

LIKE

Note the example SQL SELECT statement below shows a starting point for joining 3 tables together. The example below pertains to Week 9's practice about the TICKET, TICKET_VIOLATION, and VIOLATION tables. You do not need to JOIN 3 tables for this project (at minimum you will need to JOIN 2 tables together), but if you want to try it JOINing 3 tables together, this is an example to help you follow do that.

```
SELECT TicketDateTime, ViolationZipCode, Violation, Fine FROM TICKET
INNER JOIN TICKET_VIOLATION
ON TICKET.TicketID = TICKET_VIOLATION.TicketID
INNER JOIN VIOLATION
ON TICKET_VIOLATION.ViolationID = VIOLATION.ViolationID;
```

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RecipeID	RecipeName	Notes	AcquiredFrom	IngredientID	Ingredient	Quantity	Technique
1	Guacamole	Tasty with or without garlic.	Qdoba / trial and error	1	Avocados	10	Mash
				2	Red onion	½ large	Chopped
				3	Garlic	2 cloves	Minced
				4	Red Peppers	½ large	Chopped
				5	Limes	1	Squeezed
				6	Salt	To taste	
				7	Chips	1 Bag	
2	Grilled Pizzas	Shown with pre-made dough. Makes 12 – 9” individual pizzas	Internet (could list URL)	8	Pizza Dough	12 – 9” pre-made (or bought)	
				9	Pizza Sauce	2 cups	
				10	Olive Oil	½ cup	
				12	Pizza Cheese	8 ounces	(could buy at Fraboni's)
				13	White onion	½ large	Sliced thin.
				3	Garlic	4 tablespoons	Minced
				14	Banana Peppers	1 Jar	Pre-sliced in jar
3	Chicken Stir Fry	Can substitute vegetables as desired.		15	Chicken	2 lbs.	Thin Sliced
				4	Red Peppers	1 whole	Sliced thin, but not too thin
				16	Sweet Onion	1 Whole	Sliced, about 1.5” width

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				17	Green Beans	1 lb.	Fresh. Sliced in half
				18	Carrots	½ bag petite	Be sure they're thin.
				19	Oregano	2 teaspoons	
				20	Garlic	1 tablespoon	Minced
				21	Black Pepper	1 tablespoon	
				22	Ground Cumin	2 teaspoons	
				23	Soy Sauce	To Taste	
				24	Rice	2 cups	Could substitute with egg noodles

Project 2 Submission Instructions

Submit the following items in Brightspace:

1. Your normalized database design utilizing Crow's foot notation from Vertabelo.
2. Any assumptions or notes that are relevant to your design choices.
3. The name of the MySQL DBMS schema where your tables are implemented and the sample data was inserted.
4. One *.sql file with all of your single-table and multi-table queries.
5. Your self-evaluation and short reflection for each of the eleven project requirements specified below.

Self-evaluation and short reflection instructions:

Evaluate and reflect:

For each of the twelve project requirements below, rate your project as **Met or **Not Yet** and write out a short reflection on why you chose that rating for yourself on each project requirement.**

Note: This is not meant to be an essay. You should spend 1-2 minutes for rating and reflection of each requirement. That is a total of 12-24 minutes across all of the requirements for the written reflection portion.

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M = Meets: You have submitted evidence showing the requirement is met and you have a solid understanding of how to complete this requirement.

NY = Not Yet: You have not submitted evidence that meets this requirement and shows you have an understanding of how to complete it.

Fill in this grid as your self-rating and reflection:

Project Requirement:	Self-rating: (M or NY)	Short reflection on why you chose that rating for yourself on this project requirement:
1. Your database tables are normalized to 3NF.		
2. Your database design properly uses crow's foot notation to specify relationships between tables.		
3. Your database design has a primary keys designated properly in each table.		
4. Your database design properly enforces referential integrity utilizing foreign keys.		
5. Your database implementation matches your Vertabelo database design diagram.		
6. Your database design and implementation utilize acceptable data types for each column.		

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7. Your database design and implementation set appropriate columns as not-required (nullable).		
8. The provided sample data is properly inserted into your MySQL database implementation.		
9. Your two additional recipe examples and associated ingredients are inserted into your MySQL database implementation and they share at least one ingredient between them.		
10. Your SQL single-table SELECT queries run successfully using your MySQL database implementation.		
11. Your SQL multi-table SELECT queries run successfully using your MySQL database implementation.		

EOD – End of Document